

PROBLEM SET 3

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score: 55**)

1) **(10 pts)** Consider the coupled mass problem discussed in Sec 1.8

(a) **(5 pts)** Given that the initial state is $|x(0)\rangle = |1\rangle$, in which the first mass is displaced by unity and the second mass is left alone, calculate $|1(t)\rangle$ by following the algorithm discussed.

(b) **(5 pts)** Substitute $|x(0)\rangle = |1\rangle$ into Eq. 1.8.39 and show you get the same result as in part (a)

- 2) **(5 pts)** Consider once again the coupled mass problem discussed.
Assuming that $|\ddot{x}\rangle = \Omega|x\rangle$ has a solution $|x(t)\rangle = U(t)|x(0)\rangle$,
find the differential equation satisfied by $U(t)$.
Use the fact that $|x(0)\rangle$ is arbitrary

3) **(10 pts)** A string is displaced as follows at $t = 0$:

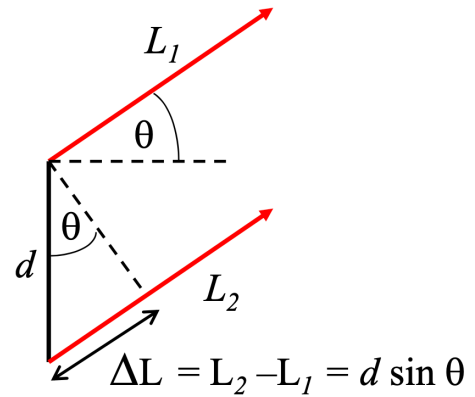
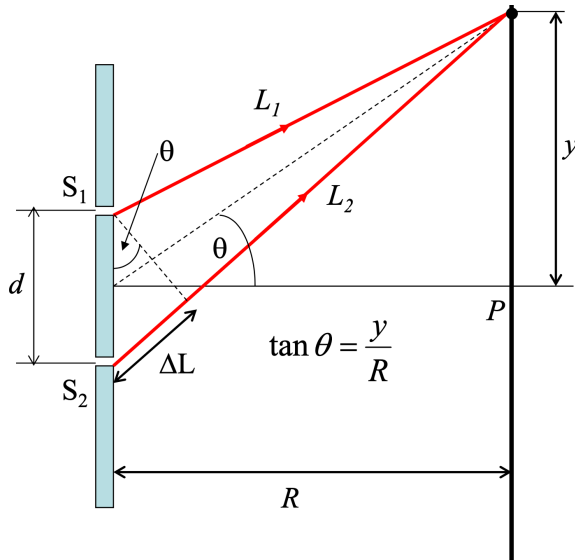
$$\psi(x,0) = \frac{2xh}{L}, \quad 0 \leq x \leq \frac{L}{2}$$

$$\psi(x,0) = \frac{2h}{L}(L - x), \quad \frac{L}{2} \leq x \leq L$$

Show that:
$$\psi(x,t) = \sum_{m=1}^{\infty} \sin\left(\frac{m\pi x}{L}\right) \cos \omega_m t \cdot \left(\frac{8h}{\pi^2 m^2}\right) \sin\left(\frac{\pi m}{2}\right)$$

Hint: Start from Eq. 1.10.55. You will end up doing two integrals, over the specified limits above.

- 4) **(30 pts)** Consider the double slit experimental set-up below, where the red lines represent the plane waves of the form: $\psi_1 = Ae^{i\phi_1}$, $\psi_2 = Ae^{i\phi_2}$. The waves pass through each slit (S_1 , S_2), where they interfere with each other before being detected at an arbitrary point y . *In the limit $R \gg d$, the lines are approximately parallel.*



- (a) **(10 pts)** Given that the intensity of each wave is defined as $I_i \equiv |\psi_i|^2$, show that the intensity due to the interference, $I = |\psi_1 + \psi_2|^2$, is given by

$$I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right),$$

where $I_0 = 4A^2$ and $\Delta\phi = \phi_2 - \phi_1$ is the phase difference between the two waves.

- (b) **(5 pts)** Derive a general expression for the location of the peaks (or bright fringes, maxima) due to constructive interference and show that it is given by

$$y_n = \frac{\lambda R}{d} n, \quad \text{where } n = 0, \pm 1, \pm 2, \dots$$

- (c) **(5 pts)** Derive a general expression for the location of the troughs (or dark fringes, minima) due to destructive interference and show that it is given by

$$y_n = \frac{\lambda R}{d} \left(n + \frac{1}{2} \right), \quad \text{where } n = 0, \pm 1, \pm 2, \dots$$

- (d) **(10 pts)** Suppose the distance between the double slits and the viewing screen is 1.2 m. The distance between the two slits is 0.030 mm. The second-order bright fringe ($n = 2$) is located 4.5 cm from the central ($n=0$) fringe.

What is the wavelength λ of the light (in nanometers) ?

What is the distance between the adjacent bright fringes (in cm) ?